

Upper Bounds on Anomalous Distributed Carrier-Phase Coherence from a GNSS Receiver Array

A pre-registered instrument program with kill switches

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Document class: White paper / pre-registered instrument program (preprint; prepared for peer-review submission)

Date: September 28, 2026 — **Revision 3.5**

Changes from Rev 3.4.

- **Title retitled toward bounds.** "Probe of vacuum structure" becomes the program's motivating question, not the paper's claim. The paper reports upper bounds; §§1–8 make no vacuum-ontology claim.
- **Ionospheric scaling corrected (§3).** 0.1 TECU gives ≈ 8.4 rad at 100 MHz vs ≈ 0.53 rad at L1 — a factor of ≈ 15.75 , not $(15.75)^2 \approx 248$. Carrier phase in radians/cycles scales as $1/f$; only delay in seconds (range in metres) scales as $1/f^2$. Rev 3.4 contradicted itself across two sentences; fixed.
- **H_C sharpened (§1, §3.1).** A phase literally identical at every antenna cancels in any inter-node difference and is unobservable by construction. H_C is redefined as anomalous phase common to all tracked carriers at each site and *correlated* across sites, with non-vanishing inter-site differential — the component that survives Chain C.
- **Double difference restored as primary Chain-S Analysis-A input (§4.0, §4.2, §5, §6).** Rev 3.4's disambiguation is reversed on reviewer advice: the 75 ps IGS clock residual (≈ 0.7 rad at L1) is identical at both nodes for a given satellite and would dominate inter-node coherence of per-node single-difference residuals. The double difference kills it exactly. Per-node post-IGS single differences are retained as a diagnostic of the subtraction, not the estimator input.
- **Estimator inputs named for both chains (§4.2, §4.3).** Analysis A and B consume two named series: Chain S — double-difference series on **disjoint node pairs**; Chain C — same-satellite inter-node difference series on **disjoint node pairs**. Detection-grade analysis requires ≥ 4 nodes; a 2–3 node deployment is a validation campaign (Phase 0), not a search.
- **Chain-C bound is a maximum (§4.4, §5).** The common-clock co-located floor does not transfer automatically to regional baselines where troposphere and multipath decorrelate. Per baseline, the reported bound is the max of the co-located non-clock floor, the isolated relative-clock contribution, and the baseline atmospheric differential RSS.
- **Common-clock transferability specified (§4.4).** Table of which terms the calibration measures directly vs which it cannot reproduce distributed; drift propagation by quadrature-added spot-check variance; deviation routes to Switch 3.

- **Switch 2: necessary, not sufficient (§7).** Troposphere, multipath, and clock error are all delay-class ($\phi_1/\phi_2 \approx 1.339$); passing rejects only ionospheric and digital-artifact explanations. Classifier is multi-component with uncertainty intervals; hardware frequency dependence bounded by the common-clock run.
- **Tracking-loop observable corrected (§4.0).** The PLL absorbs in-band phase into the NCO; the observable is NCO phase plus prompt discriminator residual, not $\text{atan2}(Q,I)$ alone. Data-bit wipeoff required on L1 C/A; cycle-slip handling pre-registered.
- **External freeze anchoring (§6).** A SHA-256 stamped by the handset clock is self-attested. The freeze is anchored externally (OSF preregistration, signed git tag, or OpenTimestamps/RFC-3161); without an anchor it is a draft, not a pre-registration.
- **Phase 0 validation campaign (§6, §8).** First campaign publishes achieved residual floor, measured L_{eff} , injection recovery, and empirical false-alarm behavior before any search claim. End-to-end blind simulations into realistic residuals are required; report empirical FAR and bound coverage.
- **Three-column results schema (§5).** Predicted budget | measured residual floor | reported bound — never let the first imply the third.
- **Appendices demoted to supplements.** Appendices A/B are excluded from the peer-review submission and maintained as separate companion notes (Supplements S1/S2), so they do not color review of the instrument work.
- **Field-book export prompts (§6).** Export prompted at every frozen milestone, not only at publication.

Instrument program: DSLV-ZPDI — the program's distributed GPS-disciplined SDR receiver array. All "array" references herein denote DSLV-ZPDI nodes unless stated otherwise.

Epistemic legend

- **[E]** Established experimental or textbook result.
- **[I]** Interpretive synthesis; not required by the data.
- **[H]** Hypothesis of this program. Contingent. Kill-switched.

Abstract

We describe a distributed array of GPS-disciplined software-defined-radio receivers (DSLV-ZPDI) that sets upper bounds on anomalous pairwise carrier-phase coherence on the GPS L1/L5 broadcast carriers after subtraction of electromagnetic propagation (Helmholtz), site multipath, clock terms, common-view GNSS errors, tropospheric delay, and ionospheric gradients. Two processing chains are distinct by construction. Chain S uses between-node double differences of between-satellite single differences; satellite and receiver clocks cancel exactly, including the ≈ 0.7 rad shared IGS clock residual at L1 that would otherwise dominate inter-node coherence. It tests direction-differential (sky-gradient) residuals (H_S) and is blind to isotropic site-common phase. Chain C uses same-satellite inter-node differences; it keeps receiver-common phase and tests correlated site-common phase with non-vanishing inter-site differential (H_C) — a literally identical global phase being unobservable by construction — with receiver clocks carried by IGS products, measured GPSDO stability, and a pre-registered common-clock calibration whose transferability is explicitly bounded.

The per-node tracking product is NCO carrier phase plus prompt-discriminator residual at dump rate f_d , with data-bit wipeoff and pre-registered cycle-slip handling. Both analyses consume two named input series on disjoint node pairs (≥ 4 nodes for detection-grade work). Analysis A bounds stationary residual coherence against a closed-form Beta null and a moving-block bootstrap; its floor is set by the effective number of independent samples, not calendar seconds. Analysis B forms a lag-resolved cross-correlation with bootstrap-max multiplicity control and lag resolution f_d^{-1} ; the $|\tau| \ll \tau_c$ morphology is claimed only on pair-of-pairs baselines with $\tau_c \gg f_d^{-1}$. Analysis A runs continuously and selects windows for deep Analysis-B follow-up; the joint false-alarm rate is calibrated end-to-end by time-slides on post-subtraction residuals and by blind injections into realistic residuals. A worked dual-frequency L-band budget shows the run is systematics-limited, with a per-snapshot double-difference phase floor of $\approx 0.4\text{--}0.8$ rad set by troposphere and multipath. This program specifies no coupling model and predicts no amplitude: it is a null experiment. Six instrument kill switches are stated, including a chromatic (uncombined L1/L5) falsification pathway that is necessary but not sufficient for any candidate excess. A null is the expected, publishable result: two upper bounds, one per chain, reported as predicted budget | measured floor | reported bound.

Keywords: distributed metrology, carrier-phase coherence, GPS-disciplined oscillators, software-defined radio, magnitude-squared coherence, null experiments, pre-registration, upper bounds

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1. The operational question

Textbook QED on Minkowski space treats the vacuum as the Poincaré-invariant ground state and localized excitations as particles. That account works in scattering and precision spectroscopy [E]. This program does not replace it, and this paper makes no vacuum-ontology claim. The instrument program DSLV-ZPDI is motivated by the vacuum-structure question; the paper's claims are upper bounds.

Operational question: after subtracting everything standard physics predicts — propagation, multipath, clocks, common-view GNSS, troposphere, ionospheric gradients, and catalogued transmitters — does a residual pairwise phase coherence remain that is inconsistent with the pre-registered null?

That question is not unique. An isotropic phase common to every tracked carrier at one antenna is cancelled by a between-satellite single difference. A satellite-common term is cancelled by a same-satellite inter-node difference. The program therefore states two observables and two bounds (§3).

Two hypotheses, named so the bounds mean something. **H_S**: an anomalous phase coupling that depends on propagation direction — a sky gradient, anisotropic vacuum structure. Tested by Chain S, which cancels isotropic site-common phase by construction. **H_C**: an anomalous phase component common to all tracked carriers at each site and **correlated across sites, with non-vanishing inter-site differential**. A phase literally identical at every antenna drops out of every inter-node difference and is unobservable by construction; H_C is the correlated differential that survives. Tested only by Chain C, which keeps receiver-common phase and therefore inherits the receiver-clock problem (§§3.1, 4.4, 5). The two chains are complementary and neither subsumes the other: a null on Chain S does not constrain H_C, and a null on Chain C does not constrain H_S. The title hypothesis lives on the harder chain. That asymmetry is the point of naming them.

This program specifies no coupling model and predicts no amplitude. It is a null experiment in the tradition of Michelson–Morley and the Holometer [E]: it bounds anomalous coherence, and a null is the expected, publishable result. If an excess survived every switch, the next step would not be an ontology paper but a quantitative signal model — amplitude, frequency dependence, baseline dependence, chain response — demonstrating that no conventional mechanism reproduces it. Data before ontology.

Carrier-phase GNSS already treats phase as the primary observable; RTK resolves cycles to millimetre precision [E]. This program extends that practice from positioning to a null search on distributed residuals. The array is a phase microscope pointed at zero.

What counts as interesting is defined in §§4–6. Everything else is scaffolding.

2. Prior art and the gap

Holometer (Fermilab). Closest interferometric prior art: two co-located 39 m Michelson interferometers, cross-spectrum to tens of MHz, including frequencies above the inverse light-crossing time of the apparatus. First published sensitivity 2.1×10^{-20} m/ $\sqrt{\text{Hz}}$ after 2×10^8 independent spectra; nulls reported on holographic shear/length noise at that floor **[E]** (Chou *et al.*, *Phys. Rev. Lett.* **117**, 111102 (2016); instrument paper arXiv:1611.08265).

Difference, stated so a reviewer does not have to reconstruct it: the Holometer is co-located optical strain at one site. This program is RF carrier-phase residuals across geographic baselines at $\sim 10^3$ MHz. A null here constrains a different observable class. It does not repeat or refute the Holometer.

Clock networks and GNSS as exotic sensors. A second line of prior art treats GNSS constellations and terrestrial clock networks as sensors for physics beyond the Standard Model. Derevianko & Pospelov proposed the GPS atomic-clock constellation as a detector of topological dark matter **[E]**, and Roberts *et al.* placed limits on domain-wall couplings using GPS satellite clock comparisons **[E]**. Those searches look for transient, propagating frequency modulation in clock comparisons. The present program's observable class — stationary, spatially distributed carrier-phase coherence after geodetic subtraction — is complementary: it requires no transient and compares phases, not frequencies. The two literatures do not constrain each other's channels, and a bound here is new information.

Methodological prior art. Morphology-agnostic detection with proper trials factors is standard practice in gravitational-wave burst searches. This program adopts the excess-power family of statistics (Anderson *et al.* 2001) and time-slide background estimation (e.g., Abadie *et al.* 2012) for Analysis B and for the joint Analysis-A $\rightarrow$ B false-alarm calibration **[E]**.

Coherence statistic, not a dynamical model. Pairwise phase coherence on a window ΔT :

$$r_{kl}(t) = \left| \frac{1}{\Delta T} \int_t^{t+\Delta T} e^{i(\phi_k(t') - \phi_l(t'))} dt' \right|.$$

Kuramoto/Strogatz are cited for the statistic only **[E]**. No coupled-oscillator dynamics are asserted for the hardware. (See §3 for the one sense in which the nodes are not independent — the GPSDO steering band — and how it is excluded.)

Phase distance $d_\varphi(a,b) = \min_n |\varphi_a - \varphi_b - 2\pi n|$ and metric distance d_g are independent observables. The empirical question **[H]** is whether $d_\varphi \approx 0$ still predicts correlated records at large d_g after every standard correlated term is removed, **on a named chain, under a named hypothesis, between two named input series.**

3. The instrument

A DSLV-ZPDI node: GNSS-disciplined oscillator, SDR, local E/B, optional barometric and radiogenic channels.

Tracked carriers. GPS broadcast carriers L1 ($f_1=1575.42$ MHz, $\lambda\approx19.0$ cm) and L5 ($f_2=1176.45$ MHz, $\lambda\approx25.5$ cm). Frequency ratio $f_1/f_2=154/115\approx1.339$. The constellation is the transmitter network; IGS final orbit/clock products are part of the null. Each node's SDR front end tracks every in-view satellite against the local GPSDO.

3.1 Two processing chains

An additive phase that is common to every tracked satellite at one antenna cancels in a between-satellite single difference. An additive phase that is common to one satellite at every antenna cancels in a same-satellite inter-node difference. The hypothesis in the title is not a single observable. Two chains are defined and pre-registered.

Chain S — sky-gradient (H_S). Between-satellite single difference at node k, same band:

$$\phi_k^{(pq)}(t) \equiv \phi_k^{(p)}(t) - \phi_k^{(q)}(t).$$

cancelling the receiver clock and any isotropic site-common phase at first order [E]. The **between-node double difference**

$$\nabla\Delta_{kl}^{(pq)} = \phi_k^{(pq)} - \phi_l^{(pq)},$$

additionally cancels satellite-clock residuals and common-view ephemeris terms exactly — including the ≈ 75 ps IGS final clock uncertainty, which is ≈ 0.7 rad at L1 and, being the *product's* error, is identical at every node using that product [E]. The double difference is the primary Analysis-A/B input for Chain S (§4.2). Chain S is **blind** to isotropic site-common phase. A null on Chain S bounds anomalous *inter-satellite differential* coherence, not isotropic common-mode phase, and does not constrain H_C.

Chain C — site-common (H_C). Same-satellite inter-node single difference, same band:

$$\Delta_{kl}^{(p)}(t) \equiv \phi_k^{(p)}(t) - \phi_l^{(p)}(t).$$

This cancels the satellite clock at first order and **keeps** receiver-common and site-common phase [E]. Receiver clocks do not cancel; they are bounded with IGS products, the measured ADEV of the installed GPSDOs, and the common-clock calibration (§4.4), whose transferability to distributed baselines is explicitly limited (§4.4, §5). Chain C is the chain that can bound correlated site-common phase with inter-site differential. It is exposed to residual receiver-clock leakage below and near the steering bandwidth (guard band, below).

A licensed ground beacon remains a possible future secondary tone; it would require its own budget rows and is out of scope.

3.2 Residuals, not raw differences

Analyses A and B consume **post-subtraction residuals** of the named chain's estimator input: IGS final orbit and clock, tropospheric hydrostatic and wet mapping, ionospheric first-order term (bound product) or the uncombined dual-frequency pair (classification product), and the site multipath phase-error model. Per-node single differences sharing the same IGS product carry the product's

clock error in common across nodes; that coherence is the null's, not a detection — which is why the double difference, not the per-node single difference, is the Chain-S estimator input. The per-node post-IGS single difference is retained as a diagnostic that the subtraction injected no structure.

3.3 Bound product and classification product

- **Bound product.** Iono-free or dual-frequency-corrected residual of the Chain-S double difference and, separately, of the Chain-C inter-node difference. This is the §5 number and the Analysis-A/B detection statistic for Switch 1.
- **Classification product.** L1 and L5 **uncombined**, same windows, same chain, same input series as any candidate. This is Switch 2. The iono-free combination mixes the two carriers and destroys the chromatic ratio; it is never the Switch 2 observable.

3.4 Timing numbers

Two timing numbers, not to be conflated:

- **Snapshot alignment.** GNSS 1 PPS / timestamp to $\ll 100$ ns so records sit on a common grid. Bookkeeping, not the science floor. At L1, 100 ns is ≈ 157 full carrier cycles.
- **Phase stability.** Short-term ADEV of the disciplined OCXO sets the reference contribution to $\phi_k^{(p)}$ before chain combinations. For $\sigma_y(\tau) \approx 10^{-12}$ at $\tau = 1$ s (mid-grade OCXO GPSDO **[E]**),

$$\sigma_\phi \approx 2\pi f_1 \tau \sigma_y \approx 9.9 \times 10^{-3} \text{ rad}$$

at $f_1 = 1575.42$ MHz. A 10^{-11} unit is ten times worse. On Chain S the receiver clock cancels in the double difference; the residual requirement enters through inter-channel biases and timestamping. On Chain C the receiver clock remains and the measured ADEV is load-bearing, with the relative-clock contribution isolated by §4.4. Quote the measured ADEV of the installed unit, not a catalog class.

Synchronization requirement, per baseline. Let $\tau_c = d_g/c$. Any claim about lag structure requires $\sigma_{\text{sync}} \ll \tau_c$, where σ_{sync} is the residual uncertainty of the time-transfer method *after* common-view correction. For 10 km ($\tau_c = 33$ μ s), common-view GNSS (ns-level) suffices. For co-located or sub-km baselines ($\tau_c \leq 1$ μ s), White Rabbit or two-way optical transfer is required **[E]**. GPS 1 PPS alignment is a simultaneity *convention*, not a measurement; the transfer method per baseline is pre-registered (§6). An excess at " $\tau \approx 0$ " is only meaningful against the stated σ_{sync} of its baseline.

Guard band: where the nodes are not independent. Every GPSDO slaves its OCXO to GPS time through a steering loop of bandwidth 10^{-3} – 10^{-2} Hz. Below that bandwidth, receiver-clock leakage is common-mode by construction. Frequencies below the measured loop bandwidth of the installed units (pre-registered per unit) are excluded from both analyses on **both chains**. The exclusion is mandatory on Chain C and conservative on Chain S. The testable frequency band is bounded below by the steering loop and above by half the prompt dump rate (§4.0); the number of independent samples within that band is bounded by the atmospheric correlation time (§4.2).

Symbol	Quantity	Unit
$\phi_k^{(p)}(t)$	carrier phase, satellite p, vs local GPSDO (NCO + discriminator)	rad

Symbol	Quantity	Unit
$\varphi_k^{(p)q}(t)$	Chain S: between-satellite single difference (diagnostic)	rad
$\nabla\Delta_{kl}^{(p)q}$	Chain S: between-node double difference (estimator input)	rad
$\Delta_{kl}^{(p)}(t)$	Chain C: same-satellite inter-node difference (estimator input)	rad
f_d	prompt-correlator dump rate	Hz
r	pairwise coherence (Analysis A)	1
$\hat{C}(\tau), \hat{r}(\tau)$	lag cross-correlation / normalized lag coherence (Analysis B)	rad ² / 1
$Q_k(t)$	quadrature residual after lock	V
τ_k	clock offset to ensemble	s
$\Gamma_{\text{ext}}(t)$	Kp, local B, pressure; optional radon	mixed

Null model, correlated terms included. Thermal noise, oscillator phase noise (residual after the chain's clock cancellation), multipath, clock residuals appropriate to the chain, common-view GNSS, tropospheric hydrostatic and wet delay, ionospheric TEC gradients, and catalogued transmitters through G_km. Dual-frequency GNSS and IGS final orbit/clock products are part of the null **[E]**. Any of these produces real inter-site coherence that is not new physics.

Band selection and the $1/f^2$ argument. Ionospheric *group* delay (and ionospheric *range* on the carrier, opposite sign) scales as $1/f^2$ in metres: $\Delta t = 40.3 \cdot \text{TEC} / (c \cdot f^2)$. Carrier phase **in radians or cycles** scales as $1/f$. A 0.1 TECU uncorrected residual — optimistic for single-frequency work — is ≈ 8.4 rad of differential *phase* at 100 MHz but ≈ 0.53 rad at L1, a factor of ≈ 15.75 . The dual-frequency iono-free combination removes the first-order term in the **bound product [E]**, leaving higher-order residuals at the millimetre level. The worked budget in §5 is L-band (L1 + L5). A sub-100 MHz carrier-phase coherence measurement across km baselines is uncorrectable with current TEC products; any band other than L-band (or higher) requires its own ionospheric error row and a spectrum-survey hash before the science run.

4. Estimators

4.0 Data path: from tracked carrier to analysis streams

The tracking loop is standard GNSS practice **[E]** (Kaplan & Hegarty), with one correction that matters: a phase-locked loop absorbs phase dynamics inside its loop bandwidth into the numerically-controlled oscillator. The per-dump observable is therefore **NCO carrier phase plus the prompt discriminator residual** — $\text{atan2}(Q, I)$ alone is the loop residual, not the carrier phase. Data-bit wipeoff is required on L1 C/A (50 bps); L5Q provides a dataless pilot. Cycle slips are detected and repaired by a pre-registered algorithm; unrepaired slips are flagged in the residual stream and excluded from Analysis-A windows.

1. **Raw.** Wideband complex IQ at sample rate f_s , held in a ring buffer (depth pre-registered; e.g. 300 s) for reprocessing and forensics.

2. **Tracking.** Per-satellite carrier and code tracking loops produce NCO phase plus prompt-discriminator I/Q dumps at dump rate f_d (pre-registered; nominal 1–10 kHz).
3. **Chain construction.** Form Chain-S single differences $\varphi_k^{(p,q)}$ and Chain-C inter-node differences $\Delta_{kl}^{(p)}$ at rate f_d . Subtract the §3 null models (IGS geometry and clocks, troposphere, ionosphere as appropriate to the product, multipath phase-error model). The post-subtraction residual of each chain is recorded continuously. For Chain S, the **double difference** $\nabla\Delta_{kl}^{(p,q)}$ is the estimator input to Analyses A and B: it cancels the satellite-clock residual exactly, including the ≈ 0.7 rad shared IGS term at L1. The per-node post-IGS single difference is a diagnostic of the subtraction, not a detection input — raw same-(p,q) single differences at two nodes are coherent through shared satellite terms and are never detection inputs.
4. **Analysis A** consumes windowed spectra of the named chain's estimator input series (§4.2). **Analysis B** consumes the same input series in the time domain (§4.3).

Dump-rate tradeoff. Coherent integration time $T=1/f_d$ sets both per-dump phase precision, $\sigma_{\varphi, \text{dump}} \approx 1/\sqrt{(2 \cdot C/N_0 \cdot T)}$, and lag resolution $\delta\tau=1/f_d$. Raising f_d sharpens the lag grid at the cost of per-dump SNR; lag-correlation integration over the trigger window recovers sensitivity per the radiometer equation. f_d is pre-registered (§6). The Analysis-B morphology test $|\tau| \ll \tau_c$ additionally requires $\tau_c \gg f_d^{-1}$ (§4.3).

A wideband residual-RF lag correlation at f_s (IQ minus reconstructed known signals) is a natural Analysis-B extension with microsecond lag resolution; it requires its own subtraction-fidelity budget row and is out of scope for this revision. Until it is in scope, regional baselines do not support a spacelike morphology claim.

4.1 Residual snapshot

On the post-subtraction residual of a named chain's estimator input (or its windowed spectrum),

$$y_k(\omega) = s_k(\omega) + n_k(\omega),$$

$$\hat{s}_k(\omega) = \sum_m G_{km}(\omega) q_m(\omega) e^{-i\omega\tau_k},$$

$$\tilde{y}_k(\omega) = (y_k(\omega) - \hat{s}_k(\omega)) e^{i\omega\tau_k}.$$

Implemented on the chain observable, the q_m are the tracked GNSS carriers, the sum runs over the satellites entering that chain product, and G_{km} is the multipath phase-error model plus geometric terms. **Definition for §§4.2–4.3:** $\tilde{y}_A, \tilde{y}_B$ denote the windowed Fourier coefficients of the two named input series of the analysis; in §4.3 the same series are used in the time domain. Chain index (S or C), product (bound or classification), and the pair-of-pairs construction are part of the series name in the archive.

4.2 Analysis A — stationary coherence (bound)

Named inputs. Coherence is computed between two input series on **disjoint node pairs**:

- **Chain S:** double-difference series $\nabla\Delta_{kl}^{(p)Q}$ and $\nabla\Delta_{mn}^{(p)Q}$, $\{k,l\} \cap \{m,n\} = \emptyset$, same satellite pair and band. All clock terms cancel exactly in each series; under the null the two series are independent.
- **Chain C:** inter-node difference series $\Delta_{kl}^{(p)}$ and $\Delta_{mn}^{(p)}$, $\{k,l\} \cap \{m,n\} = \emptyset$, same satellite. Receiver clocks are independent across the four nodes; under the null the two series are independent, and under H_C they share the correlated site-common differential.

On L non-overlapped windows:

$$\hat{r}^{(res)}(\omega) = \frac{|\sum_{t=1}^L \tilde{y}_A^{(t)} \tilde{y}_B^{(t)*}|}{\sqrt{\sum_t |\tilde{y}_A^{(t)}|^2} \sqrt{\sum_t |\tilde{y}_B^{(t)}|^2}}.$$

Magnitude-squared coherence $\hat{y} = (\hat{r}^{(res)})^2$. For windowed spectral coefficients that are complex-Gaussian under the null (asymptotic for stationary series; approximate for 1/f-type residuals — hence the bootstrap below), $\hat{y} \sim \text{Beta}(1, L-1)$ [E] (Goodman; Carter; Gish and Cochran invariance), so

$$\mathbb{E}[\hat{y}]_{\text{null}} = \frac{1}{L}, \quad \text{Var}(\hat{y})_{\text{null}} = \frac{L-1}{L^2(L+1)} \sim \frac{1}{L^2}.$$

Two qualifications: (i) the Beta law assumes non-overlapped windows and nominal degrees of freedom; overlapped windows alter the DOF and are not used in the science run; (ii) the per-window re-phasing $e^{i\omega\hat{\tau}_k}$ consumes degrees of freedom and perturbs the null at $O(1/L^2)$; the block bootstrap, not the analytic tail, is the decision statistic.

Large-L magnitude floor, used only as a check against the Beta tail:

$$\mathbb{E}[\hat{r}]_{\text{null}} \approx \sqrt{\frac{\pi}{4L}}, \quad \sigma_{\text{null}} \approx \sqrt{\frac{4-\pi}{4L}} \approx \frac{0.463}{\sqrt{L}}.$$

Independence is not free. Calendar seconds are not independent. Multipath dwell, flicker frequency noise, GNSS geometry (≈ 11.97 h repeat), and diurnal ionosphere set a correlation time τ_{corr} . Oscillator and atmospheric residuals are 1/f-type processes, for which a lag-1 autocorrelation is ≈ 1 at almost any sampling and carries almost no information about the effective sample size. Accordingly

$$L_{\text{eff}} = \frac{T}{\tau_{\text{corr}}}$$

with τ_{corr} obtained two ways on pilot data and reconciled before the science run: (a) the Politis–White automatic block-length selection for the moving-block bootstrap applied to the input series [E]; (b) an Allan-deviation-informed independence time from the measured $\sigma_y(\tau)$ turnover of the installed units [E]. The block length is then frozen in the pre-registration (§6). Detection uses L_{eff} or the bootstrap, never raw L.

Cramér–Rao phase floor on a tone, complex AWGN, N_s samples [E] (Kay 1993):

$$\text{Var}(\hat{\phi}) \geq \frac{1}{2N_s \text{SNR}}.$$

Array size. Detection-grade Analysis A requires two disjoint pairs: ≥ 4 nodes. A 2–3 node deployment executes the Phase-0 validation campaign (§6) — floor measurement, injection recovery, null characterization — and contributes single-baseline Analysis-B lag classification, but no Analysis-A detection statistic.

Multiplicity across pairs, bands, and chains. Pair-of-pairs test statistics sharing nodes are positively correlated; bands and the two chains are likewise correlated through common systematics. Benjamini–Hochberg FDR control remains valid under positive regression dependence (PRDS) [E] (Benjamini–Yekutieli 2001). Chain S and Chain C are separate families in the pre-registration; FDR is applied within family and the joint accounting is stated.

4.3 Analysis B — lag-resolved cross-correlation

Analysis B consumes the **same two named input series** as Analysis A for the chain, in the time domain at dump rate f_d , lag resolution $\delta\tau=f_d^{-1}$:

$$\hat{C}_{AB}(\tau) = \frac{1}{N} \sum_n \tilde{y}_A[n] \tilde{y}_B^*[n - \tau f_d],$$

$$\hat{r}_{AB}(\tau) = \frac{|\hat{C}_{AB}(\tau)|}{\sqrt{\hat{C}_{AA}(0) \hat{C}_{BB}(0)}}.$$

Light-travel time between the pair-of-pairs reference points (e.g. pair centroids, pre-registered): $\tau_c = D/c$.

Resolution requirement. The morphology test $|\tau| \ll \tau_c$ requires $\tau_c \gg f_d^{-1}$. Baselines not satisfying this at the pre-registered f_d are **resolution-limited** for the lag test: they contribute fully to Analysis A and to the coarse Analysis-B grid; no $|\tau| \ll \tau_c$ claim is made on them.

f_d	$\delta\tau=f_d^{-1}$	D at $\tau_c=\delta\tau$	D at $\tau_c=10\delta\tau$
1 kHz	1 ms	300 km	3000 km
10 kHz	100 μ s	30 km	300 km
IQ @ 2 MHz (deferred)	0.5 μ s	150 m	1.5 km

At nominal dump rates, a 10 km pair ($\tau_c=33 \mu$ s) and a 100 km pair ($\tau_c=333 \mu$ s) are resolution-limited for $|\tau| \ll \tau_c$. Analysis B on those baselines is a coarse-lag companion to Analysis A, not a spacelike morphology test. Raising f_d or activating the deferred IQ path extends the testable set; that choice is pre-registered, not tuned post hoc.

A 1 s Analysis-A window cannot locate a peak to better than 1 s and cannot declare a lag spacelike on any terrestrial baseline. Analysis B publishes $\hat{r}_{AB}(\tau)$ on a pre-registered τ grid containing $\{0, \pm\tau_c, \pm\tau_{iono}, \pm\tau_{trop}\}$ and a uniform span of $\pm M\tau_c$ (M pre-registered), at native resolution $\delta\tau=f_d^{-1}$.

Multiplicity control. The detection statistic is the maximum of $\hat{r}_{AB}(\tau)$ over the grid. Significance is set by a moving-block bootstrap of that maximum (block length per §4.2). No per-lag κ claims are permitted.

Trigger-selection bias and its closure. Deep Analysis-B follow-up runs on windows selected because Analysis A crossed threshold — the same noise realization. The pipeline is calibrated end-to-end: the full chain (A selection → B on selected windows) is run on time-slides — one input series shifted by unphysical offsets $\gg \tau_{\text{corr}}$ — with the slide count and the joint false-alarm rate pre-registered [E]. Slides are applied to the **post-subtraction residual streams**, so the slide background measures the background of the residual null. Correlated residuals that survive the §3 subtraction are not part of the slide background; they are the subject of Switch 1. A candidate must additionally survive a hold-out confirmation window: post-window residual data, excluded from the detection statistic, must independently show the morphology at the pre-registered level.

End-to-end validation. Before any detection claim, blind injections are run into *realistic* residuals — nonstationary, with data gaps, geometry changes, and clock steering — including null-true cases with an imperfect noise model. The program reports empirical false-positive rates and coverage of the resulting bounds, not just that the code implements the prescribed tests (§6).

Triggered architecture and data volume. Analysis A runs continuously on each chain input. Each node maintains the wideband-IQ ring buffer (§4.0) for reprocessing and forensics; when Analysis A exceeds its pre-registered trigger threshold in any band, the buffer ($\pm$ window) is dumped at all nodes simultaneously. Trigger-distribution latency ($\ll 1$ s over any terrestrial IP path) is negligible against the 300 s dwell. Continuous full-rate B recording is not required. Chain residuals are continuous and cheap (tens of kB/s per node).

Interpretation, pre-registered, **per chain, under a named hypothesis, for the named input pair:**

- Direct propagation from any single source satisfies $|\tau| \leq \tau_c$, with equality only for a source on the inter-pair axis. Peak at $\tau = \tau_c$ or a catalogued multipath extra delay: ordinary propagation. Switch 1 fails for that candidate.
- A peak at $|\tau| > \tau_c$ cannot be direct propagation. If it matches no catalogued multipath delay, it is dispersive or instrumental and triggers a pipeline investigation under Switch 3.
- Peak at $|\tau| \ll \tau_c$ (including $\tau \approx 0$ within the stated σ_{sync}), **on a pair-of-pairs satisfying $\tau_c \gg f_d^{-1}$** , that vanishes in the co-located control and vanishes when IGS/iono terms are applied: a candidate for injection-matched follow-up, subject to Switch 2, labelled with its chain and hypothesis (H_S or H_C).
- Peak at $|\tau| \ll \tau_c$ that also appears in the co-located pair-of-pairs is site-common hardware (RFI, supply, local field) unless Chain C is the intended target *and* the common-clock calibration (§4.4) has excluded the hardware reading. Co-located excess is not anomalous on Chain S.
- A common mode that is real on Chain C (H_C) would appear at $|\tau| \ll \tau_c$ in both co-located and separated pair-of-pairs, with $\delta\tau \ll \tau_c$ and $\sigma_{\text{sync}} \ll \tau_c$, would survive the common-clock decomposition in §4.4, and would pass Switch 2 in the delay class. That is the only Chain-C morphology worth a data paper, and only on baselines that are not resolution-limited.
- A $\tau \approx 0$ excess, even if every switch is passed, is common-mode noise. It is not superluminal signaling. No information transfer is possible via a common-mode field.

4.4 Injection / recovery

Before any science run: inject synthetic correlated phase at known amplitude and known lag into the raw stream; run the full pipeline on **both chains**; require recovery of amplitude and lag within pre-registered tolerance; require return to the null floor with injection off. Injections include (i) correlated site-common differential phase (must appear on Chain C and vanish on Chain S — the H_C signature) and (ii) direction-differential phase (must appear on Chain S — the H_S signature). A chain that fails its distinctive injection has no standing to report an excess.

Blind injections. A pre-registered number of injections per campaign are seeded by an injection holder who is not on the analysis team, with amplitude, lag, and chain-class drawn from pre-registered distributions and revealed only after the analysis of record is frozen. Open injections test the pipeline; blind injections test the analysts. Small-team implementation: a sealed-seed script held by a designated non-analyst; the mechanism is pre-registered, not the seed.

Common-clock calibration (breaks the Chain-C clock degeneracy). Before the science run, and repeated per campaign: one GPSDO output split to drive two complete front-end + SDR chains at a single site (common clock; independent RF paths and antennas at metre-scale separation). Chain C formed between these two front ends contains site-common phase and inter-channel biases but **no relative clock wander by construction** — it measures the non-clock systematic floor of Chain C directly, including per-channel L1/L5 inter-channel bias differences (the phase-offset-class confound for Switch 2). The independent-clock co-located pair measures site-common *plus* relative clock wander; differencing the two co-located configurations isolates the relative-clock contribution. A common-clock pair that does not sit on its predicted null routes to Switch 3 before any science claim.

Transferability: what the calibration does and does not buy. The calibration characterizes one site's hardware; it does not automatically describe separated sites.

Term	Measured directly?	Treatment
Inter-channel (L1/L5) phase biases	Yes	per-band, per-campaign; bounds the Switch-2 phase-offset confound
Front-end + splitter phase and thermal drift	Yes	at the calibration site's thermal environment
Co-located multipath / RFI / supply environment	Yes	site-specific; not transferable to other sites
Relative receiver-clock wander	Yes, by differencing the two co-located configurations	enters the Chain-C max bound
Baseline-dependent tropospheric differential	No	decorrelates with distance; budgeted per baseline from §5 rows
Far-site multipath decorrelation	No	per-site G_km; budgeted per baseline
Independent GPSDO drift between separated sites	No	bounded by per-unit ADEV plus interleaved spot checks

The calibration repeats per campaign with interleaved spot checks during long runs; measured drift variance between repeats is added in quadrature to the Chain-C floor. A spot check deviating beyond the pre-registered tolerance routes to Switch 3. **Per baseline, the reported Chain-C bound is the**

maximum of (i) the common-clock co-located non-clock floor, (ii) the isolated relative-clock contribution, and (iii) the baseline atmospheric differential RSS scaled from the §5 rows. No Chain-C number is quoted tighter than that maximum.

5. Sensitivity budget (worked example, Analysis A, L-band)

Order-of-magnitude engineering estimates, except where marked measured. Differential noise sets the per-snapshot floor σ_φ ; correlated threats must be subtracted by the null, and whatever survives is adjudicated by Switch 1. Numbers below are for the **bound product**. Every results table in the program reports three columns — **predicted budget** (this section) | **measured residual floor** (validation campaign) | **reported bound** (pre-registered analysis after systematics). Never let the first imply the third.

(a) Differential noise rows → per-snapshot σ_φ (RSS):

Parameter	Assumed value	Basis
Carriers f_0	GPS L1 1575.42 MHz + L5 1176.45 MHz	§3
Chain S observable	between-node double difference of per-node post-IGS SD	satellite <i>and</i> receiver clocks cancel exactly [E]
Chain C observable	same-satellite inter-node SD residual + IGS	satellite clock cancels; receiver clock remains [E]
Tropospheric wet-delay residual	1 cm $\Rightarrow$ ≈ 0.33 rad at L1	mapping-function and water-vapor limits [E] (Saastamoinen class)
Multipath residual, post-model	0.5–2 cm $\Rightarrow$ 0.17–0.66 rad	site-dependent; specular ground bounce [E]
Antenna PCV + cable + front-end phase	placeholder 0.1 rad; measured per unit	catalog values not accepted
Thermal noise on carrier	1 mrad	high-SNR broadcast carriers
GPSDO reference	10 mrad at 1 s before combination	installed ADEV; cancels on Chain S
Shared IGS clock residual	≈ 0.7 rad at L1 (75 ps class)	cancels exactly in the Chain-S double difference; would dominate per-node SD coherence
Chain-C clock terms	measured, per campaign	common-clock calibration (§4.4); enter the per-baseline max , not the RSS
Per-snapshot σ_φ (RSS), Chain-S DD	≈ 0.4–1.1 rad	troposphere + multipath, $\sqrt{2}$ over single difference

(b) Correlated-threat rows → null subtraction + Switch 1 (not in the RSS):

Threat	Handling
Common-view satellite orbit/clock residuals (IGS final: orbit ≈ 2 cm, clock ≈ 75 ps class)	cancelled exactly in the Chain-S double difference [E]; subtracted per-node via IGS on Chain C [E]; survival $\rightarrow$ Switch 1
Ionospheric gradients (first order removed in the bound product)	dual-frequency combination [E]; higher-order residual characterized on pilot data; survival $\rightarrow$ Switch 1
Site-common RFI / supply / local field	co-located pair-of-pairs (independent clocks, shared site, metre-scale antenna separation — mutual coupling characterized, not created; multipath similarity measured, not assumed). On Chain S this class must vanish; on Chain C it is the hypothesis class (H_C) and is separated from hardware by the common-clock calibration (§4.4), Switch 2, and Switch 6
Inter-channel (L1/L5) bias differences	calibrated per campaign in the common-clock run; the phase-offset-class confound for Switch 2
Transmitter catalog leakage	catalog version frozen; injection bounds pipeline leakage

Run parameters	
Calendar snapshots	86,400 (24 h at 1 s Analysis-A windows)
L_eff (illustrative)	10^2 – 10^3 (§4.2 on pilot data)
Detection	block-bootstrap null on $\hat{y}$, FDR (PRDS) per chain family; bootstrap-max over τ grid for B; joint FAR by time-slides on residuals; empirical FAR from blind end-to-end simulations

If one incorrectly used $L=86400$:

$$\mathbb{E}[\hat{r}]_{\text{null}} \approx 3.0 \times 10^{-3}, \quad \sigma_{\text{null}} \approx 1.6 \times 10^{-3}, \quad r_{\min}(\kappa = 3) \approx 7.7 \times 10^{-3}.$$

That number is not the program floor. For $L_{\text{eff}}=300$ ($\tau_{\text{corr}}=288$ s): $\mathbb{E}[\hat{y}]_{\text{null}}=1/300 \approx 3.3 \times 10^{-3}$, $r_{\min} \sim O(10^{-1})$. The statistical floor moves as $\sqrt{(86400/L_{\text{eff}})}$. Systematics sit above both.

Weak common phase. Two input series, shared phase φ_s (rms), independent residual phase σ_φ , treating the phase records as the measured quantity:

$$r \approx \frac{\varphi_s^2}{\varphi_s^2 + \sigma_\varphi^2} \approx \frac{\varphi_s^2}{\sigma_\varphi^2}, \quad \varphi_s \approx \sigma_\varphi \sqrt{r},$$

when $\varphi_s \ll \sigma_\varphi$. The DD floor $\sigma_\varphi \approx 0.4$ – 1.1 rad is not $\ll 1$; the millimetre conversion is an order-of-magnitude map, not an estimator.

With $\sigma_\varphi=0.42$ rad: calendar $r_{\min}=7.7 \times 10^{-3}$ gives $\varphi_s \approx 37$ mrad, ≈ 1.1 mm equivalent path at L1. With $L_{\text{eff}}=300$: $\varphi_s \approx 0.13$ rad, ≈ 4.0 mm. Report both, report the measured L_{eff} , and report Chain S and Chain C separately. The systematic floor governs.

Deliverable. Two bounds, set by different floors. **Chain S (H_S):** upper bound on anomalous sky-gradient residual phase at $O(0.1 \text{ rad})$ (mm–cm equivalent path at L1) under standard geodetic site engineering, improving only as the measured rows improve. **Chain C (H_C):** per-baseline upper bound on correlated site-common differential phase, set by the **maximum** of the common-clock co-located floor, the isolated relative-clock contribution, and the baseline atmospheric differential RSS (§4.4) — no Chain-C number is quoted tighter than that maximum. Multipath and troposphere decorrelate between nodes and inflate σ_φ rather than fake a signal; correlated terms are the danger and are handled by subtraction plus Switch 1.

6. Pre-registration and data release

Frozen before first science run, timestamped public registry. **Phase 0 is a validation campaign:** publish the achieved residual floor, measured L_{eff} , injection-recovery report, and empirical false-alarm behavior before any search claim. No detection statistic from a science run is reported without the Phase-0 null characterization on record.

- science band and spectrum-survey hash
- signal definition: tracked signals, NCO+discriminator observable, data-bit wipeoff method, cycle-slip algorithm, f_d , elevation masks, per-dump SNR characterization
- chain definitions: Chain S construction (double difference of per-node post-IGS single differences as estimator input; per-node SD as diagnostic), Chain C construction (same-satellite inter-node difference); estimator inputs as disjoint node pairs per chain; distinctive injection waveforms for H_S and H_C
- product definitions: bound product (iono-free / dual-frequency-corrected) vs classification product (L1 and L5 uncombined)
- Analysis A: window length (non-overlapped), Politis–White parameters and frozen block length, Allan cross-check, guard band f_{loop} per installed unit, selection thresholds to B, FDR procedure with PRDS clause and per-chain families, transmitter catalog version, G_{km} version per site, minimum array (≥ 4 nodes) for detection-grade analysis
- Analysis B: τ grid (M, special lags), pair-of-pairs reference-point convention and per-pair τ_c , bootstrap-max procedure, peak-location classifier, time-slide count and joint FAR target, hold-out window criteria, **resolution-limited baseline list at the chosen f_d**
- end-to-end simulation report: blind injections into realistic residuals (nonstationarity, gaps, geometry changes, steering), empirical false-positive rates, coverage of reported bounds
- IQ ring-buffer depth, dump window, multi-node dump coordination
- per-baseline time-transfer method and stated σ_{sync}
- open injection schedule, amplitudes, lags, chain-class, pass/fail windows; blind-injection count, distributions, and the identity (not the data) of the injection holder
- **common-clock calibration: splitter and front-end configuration, run schedule and interleaved spot checks per campaign, transferability table (§4.4), drift-variance propagation, predicted-null pass/fail criteria, per-band inter-channel bias calibration file hashes**

- installed GPSDO ADEV curves; measured per-node analog-chain phase vs temperature; antenna PCV calibration file hashes; co-located-pair mutual-coupling characterization
- Switch 2 pass criteria: chromatic classes with uncertainty intervals on the classification product, φ in radians or cycles; multi-component mixture policy
- exact null including §3 correlated terms, per chain, per hypothesis, per named input pair
- three-column results schema: predicted budget | measured residual floor | reported bound

Freeze anchoring. The registry freeze hash (SHA-256 over the canonical registry) is anchored externally before the science run: an OSF preregistration, a signed git tag in the public DSLV-ZPDI repository, or an OpenTimestamps / RFC-3161 timestamp. The anchor transaction ID is part of the registry. A hash stamped only by the handset clock is self-attested; without an external anchor the freeze is a draft, not a pre-registration. The field-book implementation prompts for export (JSON archive) at every frozen milestone — registry freeze, amendment re-freeze, calibration append, campaign close — not only at publication.

Raw IQ or residuals, pipeline code, injection logs (open and, after unblinding, blind), slide backgrounds, simulation reports, and the frozen pre-registration published regardless of outcome. A null is a publishable pair of upper bounds (Chain S $\rightarrow$ H_S, Chain C $\rightarrow$ H_C) at the measured L_eff and systematic floor.

7. Kill switches

Instrument switches only.

1. **Array residual (primary), per chain, per hypothesis.** Analysis A: $\hat{y}$ on the disjoint-pair input series consistent with the Beta / bootstrap null after pre-registered subtraction, FDR-controlled (PRDS), every pair-of-pairs and band, **separately for Chain S (H_S) and Chain C (H_C)**. Analysis B: the bootstrap-max over the τ grid is consistent with the time-slide-calibrated null; any excess peaks at $|\tau| \geq \tau_c$ (noting that $|\tau| > \tau_c$ without a catalogued multipath explanation routes to Switch 3), vanishes when common-view/ionospheric/IGS terms are included, or — on Chain S — is confined to the co-located control. The anomalous morphology class $|\tau| \ll \tau_c$ is claimed only on pair-of-pairs with $\tau_c \gg f_d^{-1}$. A null on Chain S is not a null on Chain C. **Blast radius:** the hypothesis **as tested by that chain**. The array remains a disciplined SDR network.
2. **Chromaticity (classification product only).** GPS L1 and L5 are separated by $f_1/f_2 \approx 1.339$. Let φ be the candidate residual **in radians or cycles**, not metres. Three classes, applied with uncertainty intervals to the **uncombined** L1/L5 pair:
 - **Delay class:** $\varphi_1/\varphi_2 = f_1/f_2 \approx 1.339$ — equal in seconds (clock, time-transfer) or metres (path). The only class that may proceed.
 - **Ionospheric class:** $\varphi_1/\varphi_2 = f_2/f_1 \approx 0.747$ — carrier-phase ionospheric residual in radians/cycles scales as $1/f$. (Ionospheric *range* in metres scales as $1/f^2$, same as group delay with opposite sign.) Routes to the null as ionosphere.
 - **Phase-offset class:** $\varphi_1/\varphi_2 = 1$ — equal in radians. Digital/processing artifact. Routes to instrumental investigation.

Passing Switch 2 is **necessary, not sufficient**: tropospheric delay, multipath, and clock error are all delay-class, so a delay-class ratio rejects only ionospheric and digital-artifact explanations. Real residuals are mixtures of these classes plus frequency-dependent hardware response and estimation error; the classifier is multi-component, hardware frequency dependence is bounded by the common-clock calibration (§4.4), and a candidate matching no single class is not promoted on chromaticity. The iono-free bound product is not used here. **Blast radius**: the candidate, not the array. This switch cannot fire on a null.

3. **Pipeline non-recovery.** Open or blind injection of known correlated phase at known lag and known chain-class is not recovered within the pre-registered window (including the distinctive S-vs-C pattern), or the off-injection state does not return to the null floor, or the time-slide background disagrees with the bootstrap null at the pre-registered level, or the end-to-end simulation's empirical FAR disagrees with the reported FAR, or the common-clock calibration pair (or an interleaved spot check) does not sit on its predicted null (non-clock floor uncharacterized, transferability broken). **Blast radius**: the campaign, until the pipeline is repaired. No science claim is permitted.
4. **Möbius holonomy as physics.** Closed RF or fiber loop of controlled area and reversed chirality yields only standard Berry / Faraday / Sagnac phases; no extra discrete π . **Blast radius**: Supplement S1 as physics. Keep as bookkeeping if useful.
5. **Gravitating plenum.** Already thrown by cosmology: $\rho_\Lambda \sim 6 \times 10^{-10} \text{ J m}^{-3}$ versus a Planck-cutoff zero-point estimate larger by tens to ~ 120 orders of magnitude depending on regularization [E]. **Blast radius**: literal Dirac-sea ontology. The phase program never needed it.
6. **Simultaneity convention.** If an Analysis-B $|\tau| \ll \tau_c$ excess is unstable under substitution of an independent time transfer (two-way optical, common-view vs all-in-view, or a second GNSS constellation), or does not survive substitution of the calibration configuration (common-clock spot check), the excess is a clock-ensemble or hardware artifact. **Blast radius**: any non-local reading of that dataset. Load-bearing on Chain C (H_C).

Aqueous QED, somatic 6.8–7.5 Hz, and Schumann coupling remain Supplement S2.

8. Closing

Delayed light is still light. A correlation is a subtraction artifact until the null model — including every *known* correlated term and the measured correlation time — fails to contain it. GPS alignment is a convention. A 1 s average is not a spacelike sample on Earth. Below the steering bandwidth of a disciplined oscillator, two GPSDOs are one clock. A between-satellite single difference is blind to isotropic site phase. A dump-rate lag axis that is coarser than d_g/c cannot test $|\tau| \ll \tau_c$. An isotropic common phase is indistinguishable from relative clock wander until a common clock says otherwise — and the common clock only speaks for its own site. A predicted budget is not a measured floor, and a measured floor is not a reported bound.

This program tests no predicted amplitude and specifies no coupling model. It sets two upper bounds on anomalous distributed RF phase coherence — Chain S on the sky gradient (H_S), Chain C on correlated site-common differential phase (H_C) — at a measured L_{eff} and measured systematic

floors, reported as predicted | measured | bound. A null is the expected result. "Systematics-limited" describes the bound, not a failure.

Run Phase 0 first: publish the achieved floor, the measured L_{eff} , the injection recoveries, and the empirical false-alarm behavior before any search claim. Build clocks good enough that phase is a spatial field. Name the chain, the hypothesis, and the two input series. Cancel the clocks each chain is allowed to cancel — exactly, in the double difference, not approximately in a product. Subtract Helmholtz, multipath, the troposphere, IGS, and the ionosphere. Reconstruct the carrier from NCO plus discriminator; wipe the data bits; flag the slips. Define the data path from tracked carrier to estimator before the first science run. Measure τ_{corr} with an estimator that respects $1/f$ noise. State σ_{sync} and $\delta\tau$ per pair-of-pairs. Split one clock across two front ends, measure Chain C's non-clock floor, bound its transferability, and take the max. Anchor the freeze outside the handset. Slide the residuals against themselves until the background is measured. Inject known lags on both chain-classes — some of them blind, into realistic residuals — and prove the pipeline finds them, puts them on the correct chain, and reports an honest false-alarm rate. Publish residuals, $\hat{r}^{\text{(res)}}$, and $\hat{r}(\tau)$. If they sit on the null, the bound is the result. If they do not, the candidate still has to survive chromaticity on the uncombined pair (necessary, not sufficient), an independent clock, a hold-out window, and a quantitative signal model that no conventional mechanism reproduces — and only then is the next paper a data paper, not another ontology.

Supplement S1. Interpretive ontology [I/H] — firewalled, not submitted

Not part of the pre-registered instrument program. Maintained as a separate companion note; nothing in §§1–7 depends on it.

S1.1. QED vacuum as a Lorentz-covariant fluctuation background whose local coherence is empirical **[H]**. Matter as a region of reduced phase coherence, not a Dirac hole (1930 hole theory is not the modern vacuum **[E]**). No operator is specified. Until one is, this is a story about the data.

S1.2. Toroidal–Möbius identification on $S^1 \times S^1$ with π holonomy per toroidal circuit; 4-cube recursion

$$C_n = (S^1 \times S^1)^{\times n} / \sim_g$$

as multi-scale phase-lock bookkeeping. No metric, no stress-energy. Switch 4 is the reading. Not LQG. Not AdS/CFT.

S1.3. Zeno as "locking," Darwinism as redundancy, Landauer as rendering cost: glosses **[I]** on established physics **[E]**. None load-bearing for the array.

S1.4. $\ell_P \approx 1.616255 \times 10^{-35}$ m. Bekenstein bound $S \leq 2\pi k_{\text{BRE}}/\hbar c$ **[E]**. Cells below ℓ_P are not extra degrees of freedom.

Supplement S2. Companion studies — gated separately, not submitted

S2.1 Somatic oscillator. BCG micromotion is real **[E]**; heart rate ~ 1 Hz. Hypothesis **[H]**: aortic–iliac mechanical peak at 6.8–7.5 Hz predicts cranial accelerometer and EEG alpha–theta coherence after heart-rate harmonics are partialled out. Control: abdominal counter-phase actuator.

S2.2 Schumann coupling. Ideal $f_1 \approx 10.6$ Hz; measured $f_1 \approx 7.8$ Hz, $f_2 \approx 14.1$ Hz, $f_3 \approx 20.3$ Hz, ambient $B \sim 0.1$ –few pT **[E]**. Hypothesis **[H]**: energetic biological entrainment. Kill: ≥ 40 dB Faraday versus Helmholtz reconstruction at ambient and $10\times$.

S2.3 Aqueous coherence. Del Giudice–Preparata–Vitiello domains (*Phys. Rev. Lett.* **61**, 1085 (1988)) are the hypothesis source **[H]**, not established fact. Kill: no collective electronic mode at $\sim 10^2$ nm, ~ 0.1 eV-class gap, 310.0 ± 0.5 K, independent labs.

Data availability

All raw IQ or residual streams, pipeline code, injection logs (open and blind, the latter after unblinding), time-slide backgrounds, simulation reports, and the frozen pre-registration will be published regardless of outcome, per §6. No part of the analysis of record may be tuned on the science data after the registry timestamp.

Competing interests

The author declares no competing financial interests. Resonant Genesis LLC funds the instrument program.

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